Remote Sensing Applications: Society and Environment 26 (2022) 100702



Contents lists available at ScienceDirect

Remote Sensing Applications: Society and Environment

journal homepage: www.elsevier.com/locate/rsase





Use of the downscalling method in satellite images for the analysis of heat islands in landfills

Atílio Grondona [a], Luciana P. Gomes [a,*], Luis M. Schiavo [a], Marcelo Caetano [a], BrunoJ.B.L. Barbosa [b]

[a] Unisinos University, Civil Engineering Post-Graduation Program, Av. Unisinos, 950, São Leopoldo, Rio Grande do Sul, 93022-750, Brazil
[b] Federal University of Rio Grande do Sul State, State Center of Research on Remote Sensing and Meteorology, Geoscience Post-Graduation Program, Av. Bento Gonçalves, 9500, Porto Alegre, Rio Grande do Sul, 91501-970, Brazil

ARTICLE INFO

Keywords:
Remote sensing
Heat islands
Landfill
Downscaling

ABSTRACT

Currently, the landfill is the most used method in the world for the treatment and final disposal of municipal solid waste (MSW), especially in first world countries. In Brazil, in 2017, only 36.8% of the municipalities used landfills for the final disposal of MSW, but this number is expected to increase significantly in the next decade. In this way, thermal infrared remote sensing (TIR) is a powerful tool for observation and monitoring the landfill. With the TIR, it is possible to evaluate various processes within the landfill, mainly those linked to biological processes that generate heat as a result. In this way, methods of obtaining the surface temperature of the landfill combined with methods of improving spatial data resolution show great potential for analyzing these data. Thus, this work uses the TIR applied to the study of environmental sanitation problems in the landfill in the city of São Leopoldo, state of Rio Grande do Sul – Brazil. The aim of this work is study the heat islands landfill using LANDSAT-8/SENTINEL-2 thermal images and consolidated downscaling methods to increase the spatial resolution of the temperature image. The results showed an increase of up to 300% in the spatial resolution of the data, allowing better qualitative discrimination of the targets, and thus identification of heat islands without increasing the root mean square error (RMSE) in temperature. In addition, the error associated with the temperature for the resampled data was less than or equal to the error associated with the original data.

1. Introduction

The alternative adopted by the human species, which grows at ever greater speeds, to live in cities, brought with it the problem of solid waste. Kaza *et al.*, 2018 reports two billion tons of solid waste generated worldwide per year. In the year of 2017, the Brazilian solid waste generation reached 60 million tons (Brasil, 2018). It is also found that the most used method in the world for the treatment and final disposal of waste is the landfill, especially in developing countries (Kaza et al., 2018), and even when the waste undergoes other treatments, such as recycling, the landfill is used to dispose of the waste coming from these processes (Butt et al., 2014; Stamps et al., 2016). In Brazil, in 2017, the disposal of MSW in sanitary landfills corresponded to only 36.8% of the municipalities, 10.9% of the municipalities sent the MSW to controlled landfills (simplified disposal system, accepted in some conditions, for example in areas that were of inadequate disposal and undergoing environmental recovery), 19.2% of the municipalities still have landfills and the rest have not responded as they proceed (Brasil, 2018).

* Corresponding author.
E-mail address: lugomes@unisinos.br (L.P. Gomes).

https://doi.org/10.1016/j.rsase.2022.100702
Received 1 November 2021; Received in revised form 22 December 2021; Accepted 27 January 2022
Available online 5 February 2022
2352-9385/© 2022 Elsevier B.V. All rights reserved.

According to data from the SNIS (National Sanitation Information System) for 2018, approximately 190 million inhabitants are served with household waste collection, with the northern region having the lowest rate (83.6%), totaling about 62.8 million tons of household and public solid waste; of which 38.1% are from selective collection with more than 27000 waste pickers involved; this entire operation has a total expense of more than R$ 22 billion reais, or about R$ 130 reais, in mean, per inhabitant (SNIS, 2019).

Due to the size of the areas involved TIR RS can be used to recover the temperature and emissivity of targets on the Earth's surface, having this information applied in several areas, such as: geology, climatology, analysis of biological processes, geophysical analysis, atmospheric plume studies, identification of soil surface constituents, land use mapping, disaster assessment, pollution detection, change detection, applications in medicine, applications in civil engineering, among many others (Collins et al., 1999, 2001).

The physical quantity measured remotely by the sensor is the spectral radiance (L_λ), defined as the radiant flow of energy in a given direction and wavelength (λ) from a normalized surface with respect to its area per unit of solid angle (Howell et al., 2015). This is valid for the entire electromagnetic spectrum (EMS), however, in the EMS regions from the visible to the medium infrared (Sobrino and Jiménez-Muñoz, 2014) the measured radiance depends solely of the target's reflective properties; in the TIR region the measured radiation depends on the target's emissivity (ε) and its temperature (T) (Kuenzer and Dech, 2013; (Rolim et al., 2016); Tang and Li, 2014). In other words, in TIR, radiance is a function of two variables (ε and T) (Rolim et al., 2016). The emissivity is an index that tells how capable a material is of transforming the absorbed energy into thermal radiation (emission) in each wavelength (Kuenzer and Dech, 2013; Tang and Li, 2014). The set of emissivities per wavelength, in the TIR portion of EMS, forms the emissivity spectrum of the target. This spectrum is very useful in the identification and characterization of materials, mainly for soils, rocks and minerals (Ninomiya, 2002; Riley and Hecker, 2013; Hecker et al., 2013).

According to Hecker et al. (2013) it is possible to classify the TIR applications in 2 groups: (1) where the emissivity spectra themselves are the main interest and are used to determine the chemical/physical parameters of the targets, something analogous to a spectroscopy; (2) where the main interest is in the temperature of the targets under study and the emissivity spectra are necessary to better determine this temperature.

Within civil engineering thermography is the analog of TIR RS, however, without the interference of the atmosphere, this because the data is obtained from a few centimeters, or meters, of the target of interest. Thus, thermography is an effective non-destructive method of testing and evaluation, successfully applied to the identification of defects and their characterization. Among the numerous applications within civil engineering, it is possible to mention: insulation studies and thermal comfort (Wild, 2007); differentiation of concrete curing stages (Palaskar, 2016); REBAR (steel reinforcing bars) test used in concrete construction to increase tensile strength (Palaskar, 2016); internal location of voids in concrete (Khan et al., 2015); structural evaluation (Hiasa et al., 2016); heat loss and thermal comfort (Wiggenhauser, 2002); identification of delamination (detachment of superficial layers of concrete resulting in a reduction in the resistance of the floor) (Clark et al., 2003); used as a non-destructive method for detecting sub-surface cracks not visible to the naked eye in secular structures, such as the gothic church of Santa Maria *ad Cryptas* built in the 13th century (Paoletti et al., 2013); search for weak points in old structures (Bisegna et al., 2014); among many others.

Still in civil engineering, more specifically in the area of environmental sanitation, TIR RS and thermography can be used to: determine the penetration of moisture into construction structures (Solla et al., 2013); (Solla et al., 2013)evaluation of plumbing and sewage in buildings and other constructions (Palaskar, 2016); and water leakage (Wild, 2007). In addition, it can be used to detect gas leaks in landfills and other pollutants, although gas emissions in landfills can be greatly reduced by various reuse processes, in practice, not all gas can be collected and some leaks can still occur (Lewis et al., 2003; Zamorano et al., 2008; Tanda et al., 2017; Fjelsted et al., 2019); for the characterization of landfill leachate through partial least squares regression in Fourier transform infrared spectroscopy (FT-IR) data (Lenz et al., 2016) or, through the same principle, in composting (Tandy et al., 2010); in the study of the local environmental impact due to the landfill (Ekeu-wei et al., 2018); for the study of bio-thermal effects in the vicinity of the landfill (Mahmood et al., 2016); in the detection of waste disposal sites using multi-temporal thermal data (Gill et al., 2019); and in the monitoring of landfills evolution with respect to their size (area) or other parameters sensitive to temperature variations (Silva et al., 2018; Iacoboaea and Petrescu, 2013).

In view of this, this work makes use of the TIR RS applied to the study of environmental sanitation problems of the landfill in the city of São Leopoldo, in the state of Rio Grande do Sul - Brazil, making a connection between geoscience and civil engineering. The objective of this work is to study the HI of the landfill using LANDSAT-8 thermal images with 100m pixel spatial resolution and consolidated downscaling methods to increase the spatial resolution of the temperature image, resampling the data to a 10m pixel of spatial resolution. The importance of studying HI in landfills is linked to the main primary products generated by the landfill (heat, gases and leachate). The HI implicitly allow to analyze the biochemical processes of decomposition of organic compounds in the landfill, since the temperature affects the decomposition of the residues, basically, in two ways: in the short term in the reaction rates and in the long term in the equilibrium of the microbial population (Yeşiller et al., 2005). In addition, high temperatures alter the mechanical and hydraulic properties of the waste, as well as the properties of the liners, roofs and surrounding soils.

2. Materials and methods

2.1. Study area

The municipality of São Leopoldo is located in bottom of the hydrographic basin of Sinos' river, belonging to the metropolitan region of Porto Alegre, 31.4 km from the capital of Rio Grande do Sul. The total area of the municipality is 102.7 km^2, of which 69.87 km^2 correspond to the urban area, 14.84 km^2 to the rural area and 17.60 km^2 to the preservation area. It has an estimated population of 238,648 inhabitants for 2020, with the last census carried out in 2010, and the HDI (human development index) of 0.739 (IBGE, 2010; São Leopoldo, 2015).

The city has a landfill operated by the Riograndense Waste Recovery Company (CRVR), located at Latitude 29.74609400 and Longitude −51.19574100. The current environmental license authorizes the operation of: a) Waste disposal cell for phases 1, 2, 3, 4, 5 and 6 (Table 1 and Fig. 1); b) Effluent treatment unit with 7 accumulation ponds; c) Rainwater emergency pond with an area of, 170 m^2; d) Administrative and auxiliary units with an area of 350.38 m^2; e) Environmental Education Center. Currently, the operation is on the upper plateau over phase 3 in yellow (Fig. 1).

The effluent generated in the waste disposal cells is taken to the accumulation ponds existing in the area, and it is not allowed to be released into the environment. The leachate accumulated in the existing lagoons in the area is sent for external treatment in companies licensed for such service. The project only accepts the receipt of MSW, Class I and Class II, and it is not allowed to receive health waste (except those previously treated for decontamination), waste from civil construction, or industrial waste. The work front of the landfill is 30 m wide and 10 m high, with the waste compacted and covered daily.

2.2. Satellite data

The LANDSAT-8-OLI (Operational Land Imager) image, acquired on April 9, 2019, has a spatial resolution of 30 m in the multi-spectral channels from the visible to the medium infrared, and 100 m in the thermal infrared channels resampled to 30m by the American Geological Service (USGS). Its radiometric resolution is 12 bits and the temporal resolution is 16 days.

The SENTINEL-2 image was acquired for the same date, April 9, 2019, with spatial resolution of 10 m in the visible and near infrared, 20 m in the shortwave and medium infrared, and 60 m in the channels dedicated to aerosol studies coastal areas, cirrus clouds and water vapor. It has radiometric resolution of 12 bits and temporal resolution of 2–10 days depending on the latitude and the number of satellites. Although the SENTINEL-2 sensor is operated by the European Space Agency (ESA), images can also be acquired without any cost from the USGS website (https://earthexplorer.usgs.gov/).

The characteristics of the LANDSAT-8/SENTINEL-2 data used for this study are summarized in Table 2.

Furthermore phases, Table 1 also shows the operation period for each phase, the total months of operation and the amount of solid waste received in each phase.

The use of thermal images to determine the surface temperature of land has been known since the 1970's (Chen et al., 2006). Although LANDSAT-8 has 2 channels in TIR, the temperature calculation must be performed with only one channel (channel 10) as the other channel (channel 11) as problems of calibration, and it is recommended not to use it to perform these calculations (USGS, 2019).

Therefore, currently there is only one thermal channel for calculating the temperature, which limits the number of methods available for this purpose. In this work, the temperature calculation was performed using the Generalized Single-channel Method (GSCM), for LANDSAT-8, proposed by Jimenez-Munoz et al. (2014). The main advantage of this method is that it does not need to directly estimate the parameters for atmospheric correction (upwelling radiation, downwelling radiation and transmittance), they are determined indirectly through the information of the water vapor content in the atmosphere.

2.3. Generalized Single-channel method (GSCM)

The calculation of the land (Earth's) surface temperature (LST) is given by (Chen et al., 2016; Jimenez-Munoz et al., 2014):

$$LST = \gamma\left[\frac{(\psi_1 L + \psi_2)}{\varepsilon} + \psi_3\right] + \delta \tag{1}$$

$$\gamma \approx \frac{T_b^2}{b_\gamma L} \tag{2}$$

$$\delta \approx T_b - \frac{T_b^2}{b_\gamma} \tag{3}$$

$$T_b \approx \frac{K_2}{ln\left(\frac{K_1}{L} + 1\right)} \tag{4}$$

$$\psi_1 = 0.04019\omega^2 + 0.02916\omega + 1.01523 \tag{5}$$

$$\psi_2 = -0.38333\omega^2 - 1.50294\omega + 0.20321 \tag{6}$$

Table 1
Grounding phases of the CRVR Landfill in São Leopoldo city.

Cell	Width (m)	Length (m)	Area (m^2)	Volume (m^3)	Period	Total Months	Total (t)
Phase 1	96	200	18.785	134.880	Dec 2011–Sep 2012	9	46,315.10
Phase 2	96	245	23.400	375.855	Oct 2012–Sep 2014	25	399,586.01
Phase 3	96	255	24.730	207.875	Oct 2014–May 2015	8	145,958.42
Phase 4	96	220	21.300	473.240	Jun 2015–Jul 2017	26	515,589.29
Phase 5	96	120	11.520	153.171	Aug 2017–Jan 2018	5	174,710.09
Phase 6	96	202	19.453	776.033	Feb 2018–Aug 2019	19	402,497.04

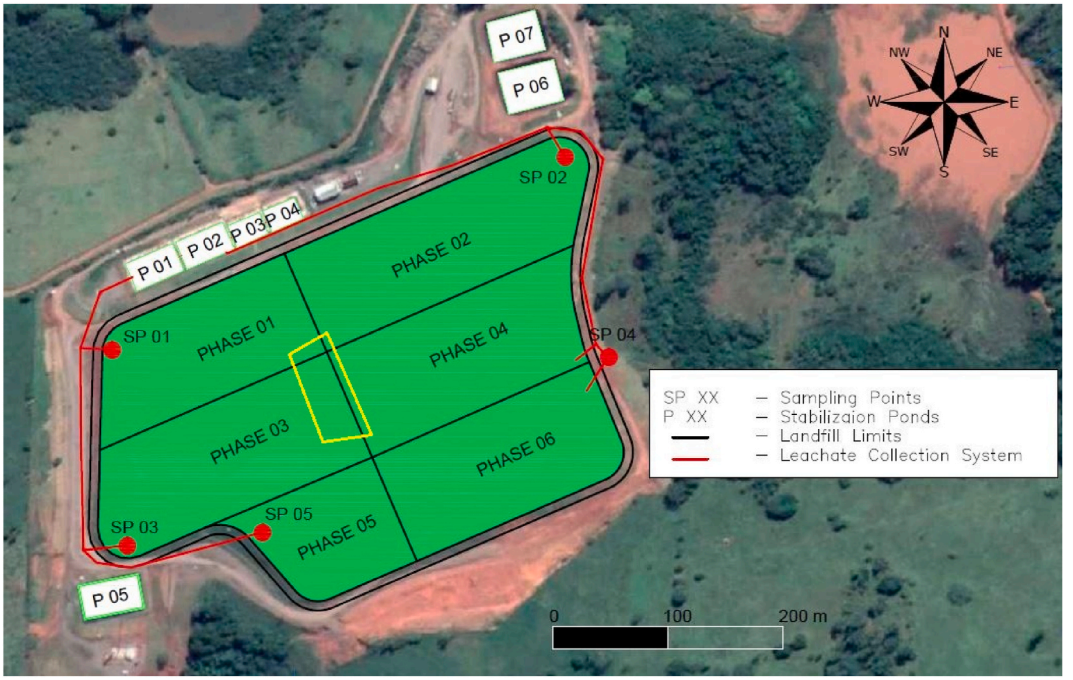


Fig. 1. Phases of the landfill operation, in yellow the location of the work front in September 2020. Adapted from: UNISINOS, 2020. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2
Details of the LANDSAT-8/SENTINEL-2 images used.

Sensor	Date of acquisition	Spatial resolution
LANDSAT-8 (OLI)	09 April 2019	30 (VIS-SWIR), 100 (TIR), 15 (PAN)[a]
SENTINEL-2	09 April 2019	10 (VIS-NIR), 20 (SWIR - MIR), 60 (Cirrus, Coastal Aerosols, Water Vapor)[a]

[a] *VIS = Visible; NIR = Near InfraRed; SWIR = Short-Wave InfraRed; MIR = Mid-Wave InfraRed; TIR = Thermal Infrared; PAN = Panchromatic.*

$$\psi_3 = 0.00918\omega^2 - 136072\omega - 0.27514 \tag{7}$$

where we have the $\gamma, \delta, \psi_1, \psi_2$ e ψ_3 being intermediate variables; ε is the pixel emissivity; K_1, K_2 e b_γ are channel-specific constants for LANDSAT-8 channel 10, with $K_1 = 774.89W/(m^2.sr.\mu m)$, $K_2 = 1321.08K$ e $b_\gamma = 1324K$. The variable T_b is the brightness temperature and L is the radiance (converted to radiance from the digital counter using ENVI 5.3 software). The term ω, given in g/cm^2, is the total water vapor content of the atmosphere, calculated using the image metadata and the FLAASH module (Fast Line-of-sight Atmospheric Analysis of Hypercubes) of the ENVI 5.3.

2.4. Emissivity calculation

The emissivity calculation from LANDSAT-8 and SENTINEL-2 images, necessary for the determination of the LST, was carried out from NDVI (Normalized Difference Vegetation Index) data as proposed by Liu and Zhang (2011) as the partial Equation (8):

$$\varepsilon = \begin{cases} 0.995\ if\ (NDVI < -0.185) \\ 0.970\ if\ (-0.185 \leq NDVI \leq 0.157) \\ 1.0094 + 0.047*\ln(NDVI) if\ (0.157 \leq NDVI \leq 0.727) \\ 0.990\ if\ (0.727 < NDVI) \end{cases} \tag{8}$$

where NDVI is given by:

$$NDVI = \frac{NIR - R}{NIR + R} \tag{9}$$

with NIR (near infrared channel) given by channel 5 for LANDSAT-8 and by channel 8 for SENTINEL-2, and R (red channel) represented by channel 4 for both sensors (LANDSAT-8/SENTINEL-2); all channels have been converted to reflectance from the digital

counter during atmospheric correction. For the SENTINEL-2 image, atmospheric correction and conversion to reflectance was performed with the SEN2COR module (https://step.esa.int/main/third-party-plugins-2/sen2cor/) of the SNAP software (Sentinel Application Platform - https://step.esa.int/main/download/), free and developed by ESA (European Space Agency) for processing SENTINEL-2 data. In the LANDSAT-8 image, the atmospheric correction and conversion to reflectance was performed with the FLAASH module of ENVI 5.3.

2.5. Calculation of albedo

According to (Vanino et al., 2018) the albedo (α) influences the net radiation of the surface, which is the primary source of energy exchange for the process of evaporation and energy balance. In this work, the surface albedo (α) was calculated from the reflectance images (visible and near infrared channels) of SENTINEL-2 sensor as proposed by Vanino et al. (2018) through a weighted average, given by:

$$\alpha = \sum_{j=1}^{n} \left| \rho_{i,j} * w_j \right| \tag{10}$$

where n is the number of channels used from SENTINEL-2; $\rho_{i,j}$ is the reflectance of the pixel i in the channel j and w_j is the weight of the channel j. The final form of Equation (11) is given by:

$$\begin{aligned} \alpha = {} & \left|0.1324 * \rho_{i,2}\right| + \left|0.1269 * \rho_{i,3}\right| + \left|0.1051 * \rho_{i,4}\right| + \left|0.0971 * \rho_{i,5}\right| + \\ & \left|0.0890 * \rho_{i,6}\right| + \left|0.0818 * \rho_{i,7}\right| + \left|0.0722 * \rho_{i,8}\right| \\ & + \left|0.0167 * \rho_{i,11}\right| + \left|0.0002 * \rho_{i,12}\right| \end{aligned} \tag{11}$$

All SENTINEL-2 channels with 20m of spatial resolution were resampled to 10m to allow the calculation of albedo.

2.6. Calculation of fractional vegetation cover (PV)

The partial vegetation cover (PV) is defined as the total vegetated percentage of the study area (roots, stems and leaves) (Zhang et al., 2019). PV does not only reflect the size of the photosynthetic area and the density of vegetation growth, but it also represents the growth trend of vegetation. In addition, it is an important parameter for the balance and development of terrestrial ecosystems and can be applied in several studies. PV was calculated from the SENTINEL-2 image as proposed by Stathopoulou and Cartalis (2009):

$$PV = \left(\frac{NDVI - mNDVI}{MNDVI - mNDVI} \right)^2 \tag{12}$$

where MNDVI and mNDVI are, respectively, the maximum and minimum NDVI value of the NDVI image of SENTINEL-2.

2.7. LST downscaling

As the LANDSAT-8 satellite has low spatial resolution in the TIR (100 m), the resulting temperature image has limitations in real world applications. Therefore, using spatial downscaling techniques, images with different spatial resolutions can be integrated, and thus used, mitigating the restrictions and limitations of the original low spatial resolution image. In this way, there is an improvement in the spatial resolution data and, consequently, in the resulting temperature image. To perform the downscaling of the LST derived from LANDSAT-8 data from SENTINEL-2 was used in the MHUTS method (Modified High-resolution Urban Thermal Sharpener) as proposed by Chen et al. (2016). The MHUTS method consists of performing a polynomial regression of grade 3 with the dependent variable being the LST, calculated from the LANDSAT-8, and as independent variables being PV, albedo (α), and the emissivity (ε), all derived from the SENTINEL-2. The NDVI from SENTINEL-2 is calculated according to Equation (8) with NIR being channel 8 and R being channel 4. The emissivity is calculated according to Table 3 from the NDVI values derived from SENTINEL-2 image. As a result of downscaling, there is an image of the LST with a spatial resolution of 10m.

According to *Chen* et al. (2016) the new LST, with a resolution of 10m, can be determined by adjusting the data using the Least Squares Method (LSM) in the following equation:

$$\begin{aligned} LST = {} & P_1.PV^3 + P_2.\alpha^3 + P_3.\varepsilon^3 + P_4.PV^2.\alpha + P_5.PV^2.\varepsilon + P_6.PV.\alpha.\varepsilon + P_7.PV.\alpha^2 + P_8.PV.\varepsilon^2 + P_9.\alpha.\varepsilon^2 + P_{10}.\varepsilon.\alpha^2 + P_{11}.PV^2 \\ & + P_{12}.\varepsilon^2 + P_{13}.\alpha^2 + P_{14}.\varepsilon.\alpha + P_{15}.PV.\alpha + P_{16}.PV.\varepsilon + P_{17}.PV + P_{18}.\alpha + P_{19}.\varepsilon + P_{20} \end{aligned} \tag{13}$$

Fig. 2 below shows the flowchart of the process with all the steps.

Table 3
Descriptive statistics of images.

	MEAN (K)	STD (K)	MAX (K)	MIN (K)	RMSE (K)
LST 30m (Fig. 3c)	298.20	2.52	307.87	279.80	- - -
LST 10m (Fig. 3d)	298.20	2.03	334.18	290.60	1.51
LST 30m (Fig. 4a)	296.03	1.62	304.44	293.42	- - -
LST 10m (Fig. 4b)	296.32	1.27	302.70	293.87	1.03

2.8. Statistical analysis

The analysis of the results was carried out in two ways: qualitatively, through visual and comparative inspection between the two images of the LST; and quantitative, by calculating the mean LST, maximum LST, minimum LST, standard deviation of the mean and RMSE (Root Mean Square Error).

According to Jensen (2000) visual inspection can be a powerful tool to extract information since the analyst use the "multi concept" and the "convergence of evidences" techniques. In this work this techniques were used in the qualitative analysis of the images. Thus, the qualitative part of the work analysis consists of visually inspecting and comparing areas with well-known behavior within the scene, such as densely vegetated areas, leachate pools, urban areas and landfill operation fronts at different scales.

In the statistical analysis, the behavior of the LST was observed by calculating the simple average for the entire scene at different scales (e.g. the image itself and a subset of the image containing the landfill), this allows finding a representative value for every scene. The maximum and minimum values (in different scales) for the LST within the scene allow to understand the amplitude of the data and how they deviate from the representative value of the scene. The standard deviation of the mean was calculated to find the error associated with the mean of the LST (on different scales), which serves to analyze how much the calculated mean varies if another sample of the area was taken for analysis. Finally, the RMSE (on different scales) was calculated, which is the measure that allows analyzing how well the model fits the data, in this measure, the closer to zero the better the model, obviously it will never be zero because, being a model, there are associated errors, however, it allows comparing the same model applied at different scales (Larson and Farber, 2019).

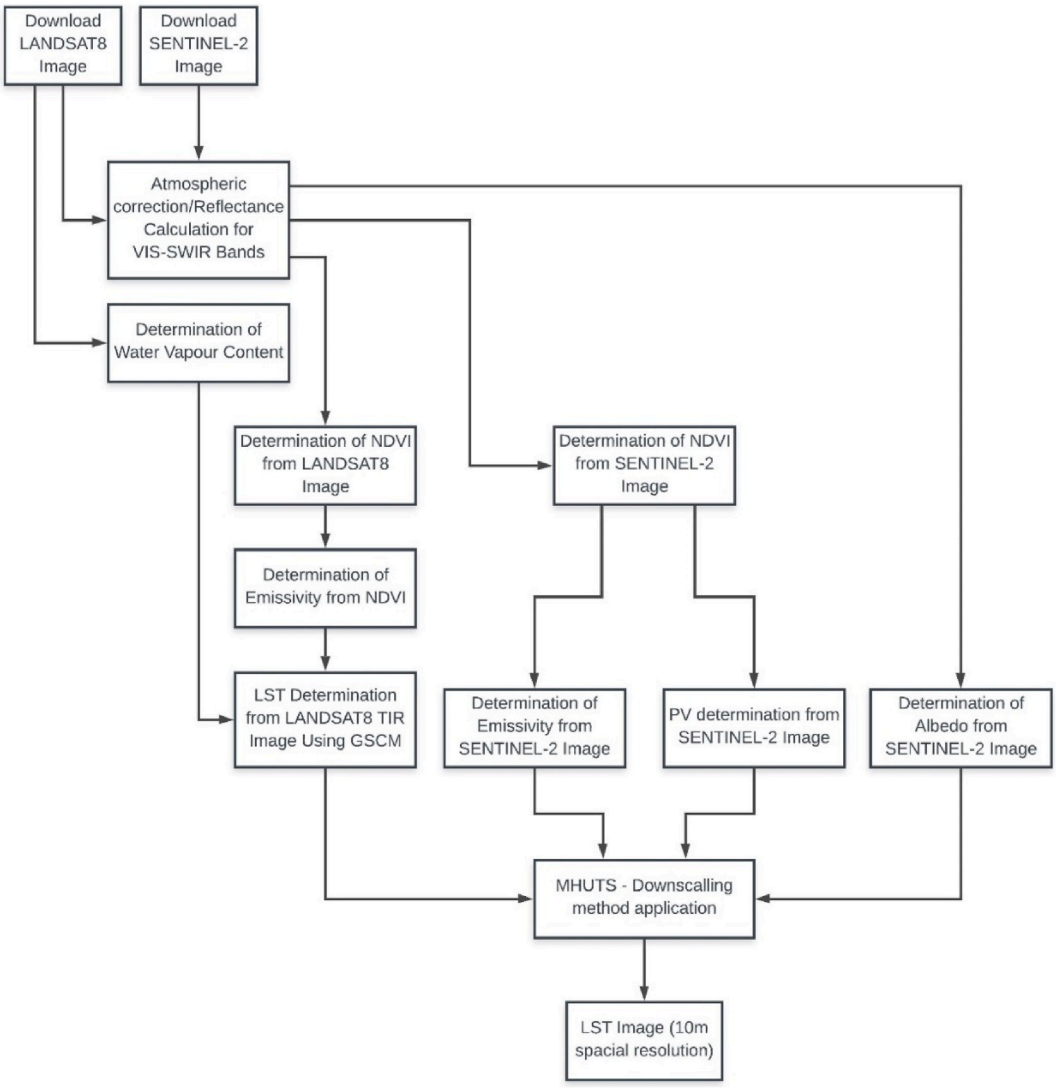


Fig. 2. Flowchart of the method used in this study.

3. Results and discussions

This work can be divided and analyzed in two parts, the determination and downscaling of LST and the analysis of heat islands. Initially, the downscaling method and its results will be discussed, and then heat islands will be analyzed.

The first step to downscaling the data is the retrieval of LST, and for this the determination of the water vapor content for the LANDSAT-8 TIR image was carried out with the FLAASH module of ENVI 5.3, resulting in 2.8861 g/cm^2. Fig. 3andb and shows, respectively, the LANSDSAT-8 and SENTINEL-2 red channel images corrected for atmospheric distortion, only for comparison of spatial resolution. It should be notice the visual quality gain (improvement) when using the sensor with higher spatial resolution (Fig. 3b).

The emissivity, PV and albedo images are visually very similar among then and similar to Fig. 3a, for this reason we have omitted these images in this work. As expected, the maximum albedo was close to 0.47 which is in accordance with the average earth albedo

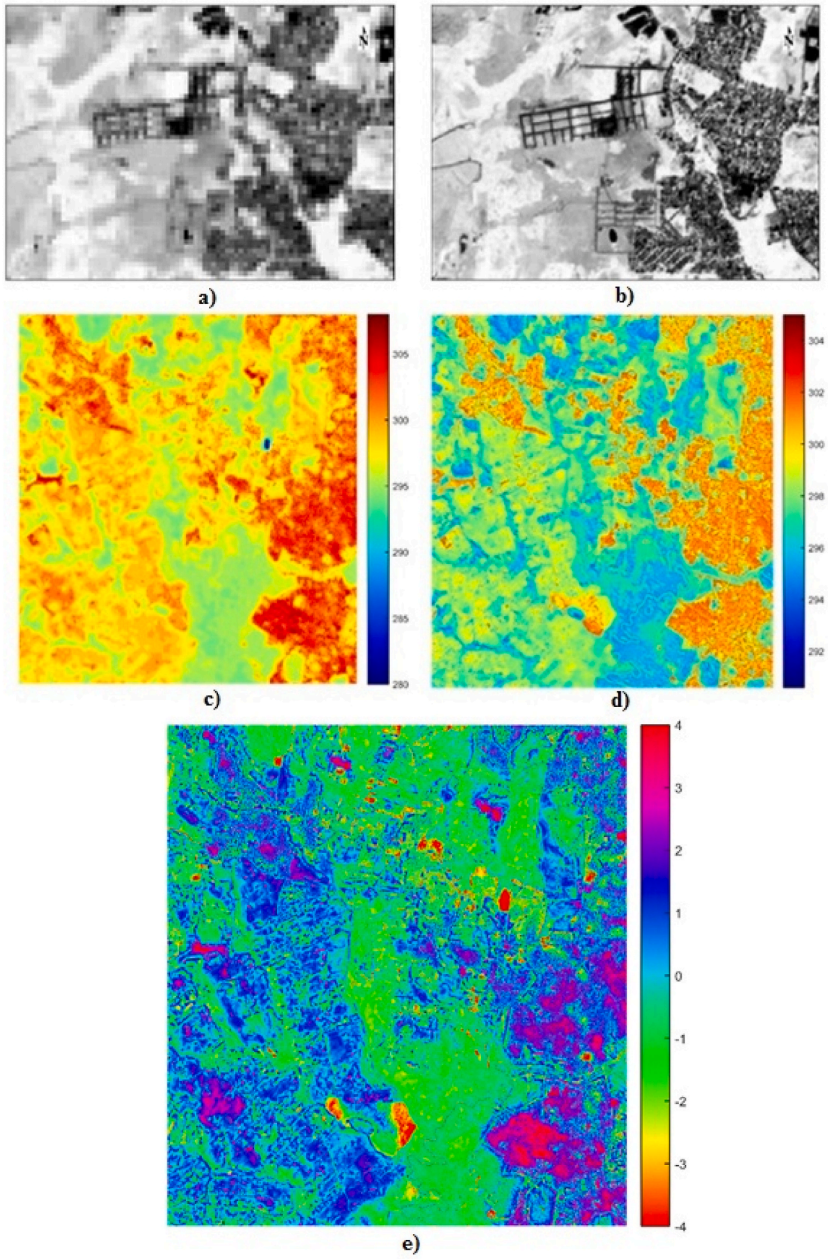


Fig. 3. a) Red channel of a subset area of LANDSAT-8 image (30 m); b) Red channel of a subset area of SENTINEL-2 image (10 m); c) LST image (30 m) calculated with the GSCM method from the LANDSAT-8 image; d) LST image (10 m) calculated with MHUTS; e) Image difference between the LST with 30m and the LST with 10m (image c – image d).

which is approximately 0.4.

From the information of water vapor content, the NDVI data, and the emissivity data of LANDSAT-8 it was possible to calculate the LST using the GSCM method (Fig. 3c). The calculated LST has a spatial resolution of 30 m and is given in Kelvin (K). In Fig. 3d it is possible to see the LST with a resolution of 10 m (downscaling), where the improvement in the visual quality of the image is noticeable, as well as in the discriminating power over the targets. Thus, there was a significant improvement in the visual quality of the new LST due to the increase in spatial resolution caused by the downscaling algorithm.

In Fig. 3e there are areas such as vegetation and soil had their temperatures underestimated (areas in green, yellow and red) by MHUTS.

Table 3 presents a summary of the descriptive statistics of the images (full and subset images) for the original images and for the downscaling images. The average temperature value for both full images (Fig. 3and d) remained unchanged, however the standard deviation of the mean LST with 10 m resolution decreased by 0.49 K, which is equivalent to an error 19.44% smaller. The temperature range went from ≈28 K (in 30 m) to ≈43 K (in 10 m), and this change occurs mainly due to the adjustment made with the least squared method. The RMSE for the 10 m LST was 1.51 K, and there is no RMSE for the 30 m LST image because no one adjustment was performed in these data.

The last two rows of Table 3 (Fig. 4a and b rows) also presents a summary of the descriptive statistics for the subset of the images. The averages temperatures values will result in a difference <0.3K. The standard deviation of the 10 m LST was 0.35 K less than the 30 m LST, which is equivalent to an error 21.6% smaller. Finally, the data range went from 11.02 K (in 30 m image) to 8.83 K (in 10 m image).

It is evident that the smaller the subset of the study area is, the smaller the RMSE will be, which is expected since the variability of the data is reduced by reducing the observed area. In addition, the subset of the study area excludes problematic targets such as dense vegetation areas and urban regions, which are the main causes of the under/overestimation of the temperature with the downscaling method. It was also observed that the MHUTS underestimates the data in some regions, but the underestimated value is in the same order of magnitude as the RMSE. Qualitatively, the increase in the discriminating power that MHUTS adds is undeniable, even though there is a slight variation in temperature resampled. These variations in the calculations of temperature are of the same order of magnitude than the radiometric resolution of the sensor.

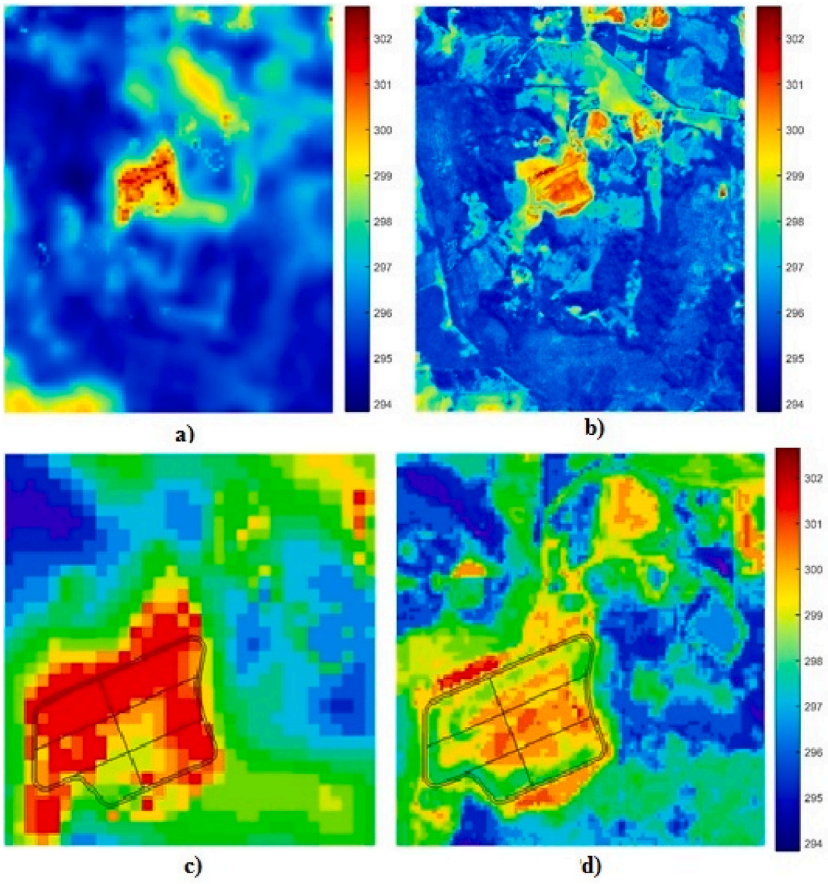


Fig. 4. a) LST image subset over landfill's operating area (30 m) calculated with the GSCM; b) LST image subset over landfill's operating area (10 m) calculated with MHUTS; c) Zoom in of image a) and phases over landfill's operating area (30 m); d) Zoom in of image b) and phases over landfill's operating area (10 m).

As the objective of the work is to specifically analyze the landfill's operating region (Fig. 1), MHUTS was applied a second time in a subset of the image, only in the area of interest (Fig. 4). This resulted in a 0.48 K decrease in RMSE (Table 3, Fig. 4a and b rows), or an improvement of up to 31.79% in the data adjustment if compared to the whole image.

Analyzing qualitatively the results it is possible to see in Fig. 3c regions of high temperature, these regions are basically urban areas (≈305K) and exposed soils (≈302K), with urban areas being predominant. A blue spot can also be seen in the image, indicating a low temperature region (≈280K), this spot is a cloud present in the image, and as expected its temperature is lower than the Earth's mean temperature. The vegetated area and the rivers have temperatures close to 295 K. It is worth mentioning that even if the LST is accurate the detailed analysis of the targets is compromised due to the low spatial resolution (30 m) of the product.

As in Fig. 3d, the areas with the highest temperature are urban areas (≈303K) and exposed soils (≈300K), followed by water bodies (≈297K) and dense vegetation (≈293K). In Fig. 3d, it can also be seen that within the urban area there was a good identification of the vegetated areas, which does not occur in Fig. 3c.

The areas in blue, pink and red (Fig. 3e) had their temperatures overestimated, being these areas predominantly corresponding to urban spots and buildings. Thus, the downscaling algorithm has a temperature range of 10 K (±5 K), however the area of interest in this study, which is the landfill, has a lower temperature range of 2 K (±1 K). The fact that this variation is symmetrical in amplitude around "0" gives it characteristics of randomness, implying a good adaptation of the model used.

The downscaling method applied in the subset LST image with 10 m (Fig. 4b) presents a better visualization of details in temperature image, having superior target discriminating power than the LST of 30 m (Fig. 4a), and with a less variation in the adjusted values of the regression. Fig. 4c is a zoom in from Fig. 4a, with the landfill area superimposed in black showing the different phases of operation in the landfill (Fig. 1). In Fig. 4d the leachate ponds are well defined (red) while there are temperature peaks in phase 4 (red-orange), followed by HI in phases 3 and 4 (dark orange), both expanding for phases 5 and 6. This shows that the applied method was able to identify hot spots and their possible trajectory towards the current work front which is in phase 6. A possible cause of the current work front not have the highest temperature may be because it is not covered yet and can be a loss of heat due to wind, rain, and other external factors. In Fig. 4b we have the same area but with little descriptive power over the work fronts due to the low spatial resolution (30 m), which prevents further analysis using this data because the entire area of the landfill is presented as a large HI (Red).

4. Conclusion

The TIR RS can be used to recover the temperature and emissivity of targets on the Earth's surface, with this information being applied in several areas. Many areas are interested in determining the emissivity spectrum for the study, identification, and classification of targets; while others focus on determining the temperature. However, independently of the application this is an ill-posed problem (more variables than equations), since the measured radiance is a function of the object's temperature and its emissivity. Thus, it is necessary to know the temperature to determine the emissivities or vice-versa. There are many applications of TIR RS for Civil Engineering, and an analogy can be made with thermography, however on a larger scale. With this in mind, this work aimed to study the HI in a landfill in the city of São Leopoldo - RS, comparing the results obtained directly from medium resolution orbital sensors such as LANDSAT-8 with the results obtained from algorithms of downscaling. For this study, the MHUTS downscaling algorithm was applied to the 30m temperature image to obtain a 10 m resampled temperature image. There was an increase of 300% in the spatial resolution of the LST, calculated from the LANDSAT-8 sensor, going from a spatial resolution of 30 m–10 m. This increase in resolution allowed a better qualitative discrimination of the targets of the study area, also allowing better discrimination of hot spots within the area of interest. The LST with 10 m of spatial resolution has RMSE <1.154 which is in accordance with Chen et al. (2016). In addition, for the subset of the area of interest, the amplitude and standard deviation of the mean LST with a resolution of 30 m was greater than for the LST with a resolution of 10 m, in other words, the errors of the LST resampled using the downscaling method are equal or lower than the original LST. Thus, it is possible to say that the downscaling method allows an improvement in details to be made without a significant loss in the accuracy of the LST.

For future research, this algorithm will be applied a landfill that is currently in the process of discontinuing the operation, and migrating the operation to an adjacent area, in a new landfill that is currently under construction. In this way, the algorithm proposed in this work will be used in a landfill completely closed, allowing to monitoring it and analyze the relationship between the stability of MSW and heat emissions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) - Finance Code 001.

References

Bisegna, F., Ambrosini, D., Paoletti, D., Sfarra, S., Gugliermetti, F., 2014. A qualitative method for combining thermal imprints to emerging weak points of ancient wall structures by passive infrared thermography – a case study. J. Cult. Herit. 15, 199–202. https://doi.org/10.1016/j.culher.2013.03.006.

Brasil, 2018. Ministério das Cidades, Secretaria Nacional de Saneamento Ambiental. Diagnóstico do Manejo de Resíduos Sólidos Urbanos 2017. Brasília, Sistema Nacional de Informações sobre Saneamento - SNIS. http://www.snis.gov.br/diagnostico-residuos-solidos/diagnostico-rs-2017 (accessed in 04 april 2019).
Butt, T.E., Gouda, H.M., Baloch, M.I., Paul, P., Javadi, A.A., Alam, A., 2014. Literature review of baseline study for risk analysis — the landfill leachate case. Environ. Int. 63, 149–162. https://doi.org/10.1016/j.envint.2013.09.015.
Chen, J., Ding, F., Li, Q., Wu, W., Fan, P., Zhang, Xin, 2016. Using a modified HUTS algorithm to downscale Land Surface Temperature retrieved from Landsat-8 imagery: a case study of Xiamen City, China. In: 2016 4th International Workshop on Earth Observation and Remote Sensing Applications (EORSA). IEEE, pp. 38–42. https://doi.org/10.1109/EORSA.2016.7552762.
Chen, X.-L., Zhao, H.-M., Li, P.-X., Yin, Z.-Y., 2006. Remote sensing image-based analysis of the relationship between urban heat island and land use/cover changes. Remote Sens. Environ. 104, 133–146. https://doi.org/10.1016/j.rse.2005.11.016.
Clark, M., McCann, D., Forde, M., 2003. Application of infrared thermography to the non-destructive testing of concrete and masonry bridges. NDT E Int. 36, 265–275. https://doi.org/10.1016/S0963-8695(02)00060-9.
Collins, E.F., Roberts, D.A., Borel, C.C., 2001. Spectral mixture analysis of simulated thermal infrared spectrometry data: an initial temperature estimate bounded TESSMA search approach. IEEE Trans. Geosci. Rem. Sens. 39, 1435–1446. https://doi.org/10.1109/36.934075.
Collins, E.F., Roberts, D.A., Sutton, P.C., Funk, C.C., Borel, C.C., 1999. In: Descour, M.R., Shen, S.S. (Eds.), title>Temperature Estimation and Compositional Mapping Using Spectral Mixture Analysis of Thermal Imaging Spectrometry Data</title>, pp. 286–297. https://doi.org/10.1117/12.366291.
Ekeu-wei, I.T., Azuma, K.I., Ogunmuyiwa, F.B.B., 2018. Assessment of environmental impact of solid waste dumpsites using remote sensing. Niger. J. Technol. 37, 275. https://doi.org/10.4314/njt.v37i1.36.
Fjelsted, L., Christensen, A.G., Larsen, J.E., Kjeldsen, P., Scheutz, C., 2019. Assessment of a landfill methane emission screening method using an unmanned aerial vehicle mounted thermal infrared camera – a field study. Waste Manag. 87, 893–904. https://doi.org/10.1016/j.wasman.2018.05.031.
Gill, J., Faisal, K., Shaker, A., Yan, W.Y., 2019. Detection of waste dumping locations in landfill using multi-temporal Landsat thermal images. Waste Manag. Res. 37, 386–393. https://doi.org/10.1177/0734242X18821808.
Hecker, C.A., Smith, T.E.L., da Luz, B.R., Wooster, M.J., 2013. Thermal Infrared Spectroscopy in the Laboratory and Field in Support of Land Surface Remote Sensing, pp. 43–67. https://doi.org/10.1007/978-94-007-6639-6_3.
Hiasa, S., Birgul, R., Catbas, F.N., 2016. Infrared thermography for civil structural assessment: demonstrations with laboratory and field studies. J. Civ. Struct. Heal. Monit. 6, 619–636. https://doi.org/10.1007/s13349-016-0180-9.
Howell, J.R., Menguc, M.P., Siegel, R., 2015. Thermal Radiation Heat Transfer, sixth ed. CRC Press. https://doi.org/10.1201/b18835.
Iacoboaea, C., Petrescu, F., 2013. Landfill monitoring using remote sensing: a case study of Glina, Romania. Waste Manag. Res. 31, 1075–1080. https://doi.org/10.1177/0734242X13487585.
IBGE - Instituto Brasileiro de Geografia e Estatística. Cidades - São Leopoldo, 2010. https://www.ibge.gov.br/cidades-e-estados/rs/sao-leopoldo.html (accessed 08 dezember 2020).
Jimenez-Munoz, J.C., Sobrino, J.A., Skokovic, D., Mattar, C., Cristobal, J., 2014. Land surface temperature retrieval methods from landsat-8 thermal infrared sensor data. Geosci. Rem. Sens. Lett. IEEE 11, 1840–1843. https://doi.org/10.1109/LGRS.2014.2312032.
Jensen, J.R., 2000. Remote Sensing of the Environment: an Earth Resource Perspective.
Kaza, S., Yao, L., Bhada-Tata, P., Woerden V., F., et al., 2018. What a Waste 2.0: A Global Snapshot of Solid Waste Management to 2050. Washington: World Bank. https://openknowledge.worldbank.org/handle/10986/30317, 2018. (Accessed 4 July 2019).
Khan, F., Bolhassani, M., Kontsos, A., Hamid, A., Bartoli, I., 2015. Modeling and experimental implementation of infrared thermography on concrete masonry structures. Infrared Phys. Technol. 69, 228–237. https://doi.org/10.1016/j.infrared.2015.02.001.
Kuenzer, C., Dech, S. (Eds.), 2013. Thermal Infrared Remote Sensing, Remote Sensing and Digital Image Processing. Springer Netherlands, Dordrecht. https://doi.org/10.1007/978-94-007-6639-6.
Larson, R., Farber, E., 2019. Elementary Statistics: Picturing the World.
Lenz, S., Böhm, K., Ottner, R., Huber-Humer, M., 2016. Determination of leachate compounds relevant for landfill aftercare using FT-IR spectroscopy. Waste Manag. 55, 321–329. https://doi.org/10.1016/j.wasman.2016.02.034.
Lewis, A.W., Yuen, S.T.S., Smith, A.J.R., 2003. Detection of gas leakage from landfills using infrared thermography - applicability and limitations. Waste Manag. Res. 21, 436–447. https://doi.org/10.1177/0734242X0302100506.
Liu, L., Zhang, Y., 2011. Urban heat island analysis using the landsat tm data and aster data: a case study in Hong Kong. Rem. Sens. 3, 1535–1552. https://doi.org/10.3390/rs3071535.
Mahmood, K., Batool, S.A., Chaudhry, M.N., 2016. Studying bio-thermal effects at and around MSW dumps using Satellite Remote Sensing and GIS. Waste Manag. 55, 118–128. https://doi.org/10.1016/j.wasman.2016.04.020.
Ninomiya, Y., 2002. In: Maldague, X.P., Rozlosnik, A.E. (Eds.), Mapping Quartz, Carbonate Minerals, and Mafic-Ultramafic Rocks Using Remotely Sensed Multispectral Thermal Infrared ASTER Data, pp. 191–202. https://doi.org/10.1117/12.459566.
Palaskar, S.M., 2016. Application OF infrared thermography IN civil engineering. Int. J. Adv. Technol. Eng. Sci. 4, 8.
Paoletti, D., Ambrosini, D., Sfarra, S., Bisegna, F., 2013. Preventive thermographic diagnosis of historical buildings for consolidation. J. Cult. Herit. 14, 116–121. https://doi.org/10.1016/j.culher.2012.05.005.
Riley, D.N., Hecker, C.A., 2013. Mineral Mapping with Airborne Hyperspectral Thermal Infrared Remote Sensing at Cuprite. Nevada, USA, pp. 495–514. https://doi.org/10.1007/978-94-007-6639-6_24.
Rolim, S.B.A., Grondona, A., Hackmann, C.L., Rocha, C., 2016. A review of temperature and emissivity retrieval methods: applications and restrictions. Am. J. Environ. Eng. 6, 9. https://doi.org/10.5923/s.ajee.201601.18.
São Leopoldo, 2015. Localização. http://www.saoleopoldo.rs.gov.br/ (accessed 10 september 2019).
Silva, J.F., Silva, R.K.A. da, Paz, Y.M., Reis, J.V., dos, Ferreira, H.D.S., Candeias, A.L.B., 2018. Avaliação do sensoriamento remoto termal para monitoramento de aterros sanitários. J. Environ. Anal. Prog. 3 https://doi.org/10.24221/jeap.3.1.2018.1536.37-48, 037.
SNIS - Sistema Nacional de Informações sobre Saneamento, 2019. Painel resumo dos dados do manejo dos resíduos sólidos urbanos para o ano de 2019. http://www.snis.gov.br/painel-informacoes-saneamento-brasil/web/painel-residuos-solidos(accessed 15 january 2021).
Sobrino, J.A., Jiménez-Muñoz, J.C., 2014. Introduction to optical/thermal remote sensing. URL. http://seom.esa.int/landtraining2014/files/D1T1a_Sobrino-Jimenez.pdf.
Solla, M., Lagüela, S., Riveiro, B., Lorenzo, H., 2013. Non-destructive testing for the analysis of moisture in the masonry arch bridge of Lubians (Spain). Struct. Control Health Monit. https://doi.org/10.1002/stc.1545.
Stamps, B.W., Lyles, C.N., Suflita, J.M., Masoner, J.R., Cozzarelli, I.M., Kolpin, D.W., Stevenson, B.S., 2016. Municipal solid waste landfills harbor distinct microbiomes. Front. Microbiol. 7 https://doi.org/10.3389/fmicb.2016.00534.
Stathopoulou, M., Cartalis, C., 2009. Downscaling AVHRR land surface temperatures for improved surface urban heat island intensity estimation. Remote Sens. Environ. 113, 2592–2605. https://doi.org/10.1016/j.rse.2009.07.017.
Tanda, G., Migliazzi, M., Chiarabini, V., Cinquetti, P., 2017. Application of close-range aerial infrared thermography to detect landfill gas emissions: a case study. J. Phys. Conf. Ser. 796, 012016 https://doi.org/10.1088/1742-6596/796/1/012016.
Tandy, S., Healey, J.R., Nason, M.A., Williamson, J.C., Jones, D.L., Thain, S.C., 2010. FT-IR as an alternative method for measuring chemical properties during composting. Bioresour. Technol. 101, 5431–5436. https://doi.org/10.1016/j.biortech.2010.02.033.
Tang, H., Li, Z.-L., 2014. Quantitative Remote Sensing in Thermal Infrared. Springer Remote Sensing/Photogrammetry. Springer Berlin Heidelberg, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-42027-6.
UNISINOS - Universidade do Vale do Rio dos Sinos. 2020. Relatório 11 - Consultoria Técnica Sobre Resultados do Monitoramento Ambiental do Aterro Sanitário da CRVR em São Leopoldo. São Leopoldo - RS 60, 2020, 2020.
USGS, 2019. Landsat 8 OLI and TIRS calibration notices. URL. https://www.usgs.gov/land-resources/nli/landsat/landsat-8-oli-and-tirs-calibration-notices.

Vanino, S., Nino, P., De Michele, C., Falanga Bolognesi, S., D'Urso, G., Di Bene, C., Pennelli, B., Vuolo, F., Farina, R., Pulighe, G., Napoli, R., 2018. Capability of Sentinel-2 data for estimating maximum evapotranspiration and irrigation requirements for tomato crop in Central Italy. Remote Sens. Environ. 215, 452–470. https://doi.org/10.1016/j.rse.2018.06.035.

Wiggenhauser, H., 2002. Active IR-applications in civil engineering. Infrared Phys. Technol. 43, 233–238. https://doi.org/10.1016/S1350-4495(02)00145-7.

Wild, W., 2007. Application of infrared thermography in civil engineering. Proc. Est. Acad. Sci. Eng. 13, 436–444.

Yeşiller, N., Hanson, J.L., Liu, W.-L., 2005. Heat generation in municipal solid waste landfills. J. Geotech. Geoenviron. Eng. 131, 1330–1344, 10.1061/ASCE)1090-0241, 2005.

Zamorano, M., Brebbia, C.A., Kungolos, A.G., Popov, V., Itoh, H., 2008. Waste Management and the Environment IV. Wit, Southampton.

Zhang, S., Chen, H., Fu, Y., Niu, H., Yang, Y., Zhang, B., 2019. Fractional vegetation cover estimation of different vegetation types in the qaidam basin. Sustainability 11, 864. https://doi.org/10.3390/su11030864.